MCT 211
Automatic Control

Stability Analysis (Routh Criterion)

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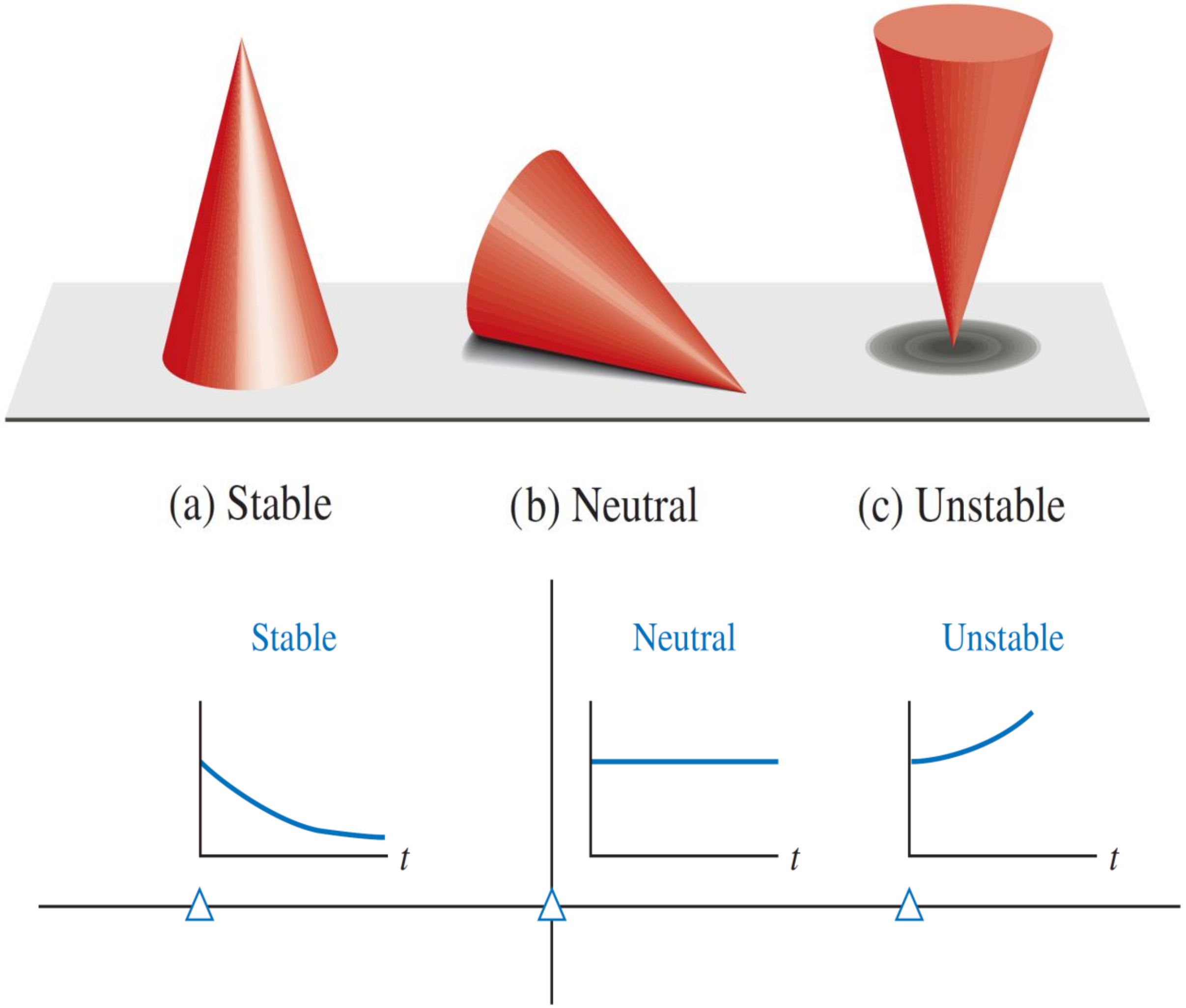
Stability Analysis

Bounded-input, Bounded-output stability

A **stable system** is defined as a system with a ***bounded*** (limited) system response. That is, if the system is subjected to a **bounded input** or **disturbance** and the **response** is ***bounded*** in magnitude, the system is said to be **stable**.



Stability Analysis

Bounded-input, Bounded-output stability

A **necessary** and **sufficient condition** for a feedback system to be **stable** is that **all** the **poles** of the system transfer function have **negative real parts**.

$$T(s) = \frac{p(s)}{q(s)} = \frac{K\prod_{i=1}^{M}(s + z_i)}{s^N \prod_{k=1}^{Q}(s + \sigma_k)\prod_{m=1}^{R}\left[s^2 + 2\alpha_m s + \left(\alpha_m^2 + \omega_m^2\right)\right]}$$

where $q(s) = \Delta(s) = 0$ is the characteristic equation, whose roots are the poles of the closed-loop system.

Stability Analysis

Bounded-input, Bounded-output stability

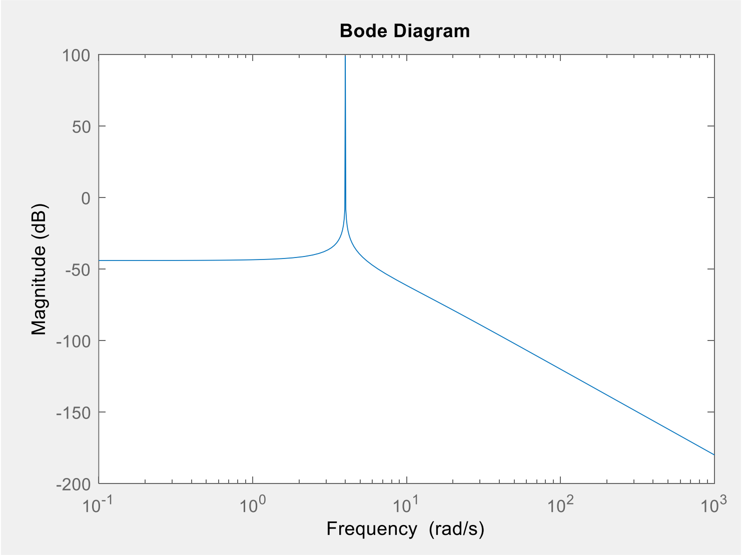
- A system is **stable** if **all the poles** of the transfer function are in the **left-hand** s-plane.
- A system is **not stable** if **not all** the **roots** are in the **left-hand** plane.
- If the characteristic equation has simple roots on the imaginary axis (jω-axis) with all other roots in the left half-plane, the steady-state output will be sustained oscillations for a bounded input.
- For an **unstable system**, the characteristic equation has at least one **root** in the **right half** of the s-plane or **repeated** ***jω*** **roots**; for this case, the output will become **unbounded** for any input.

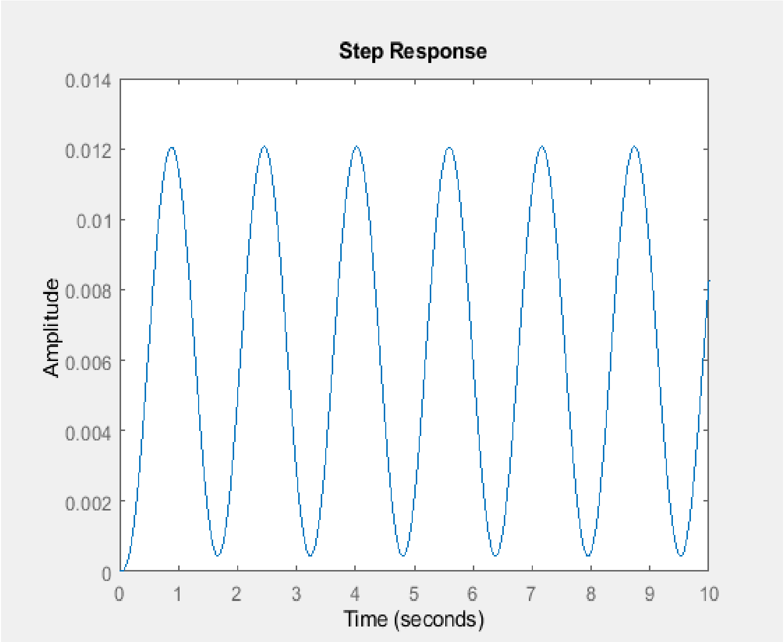
Stability Analysis

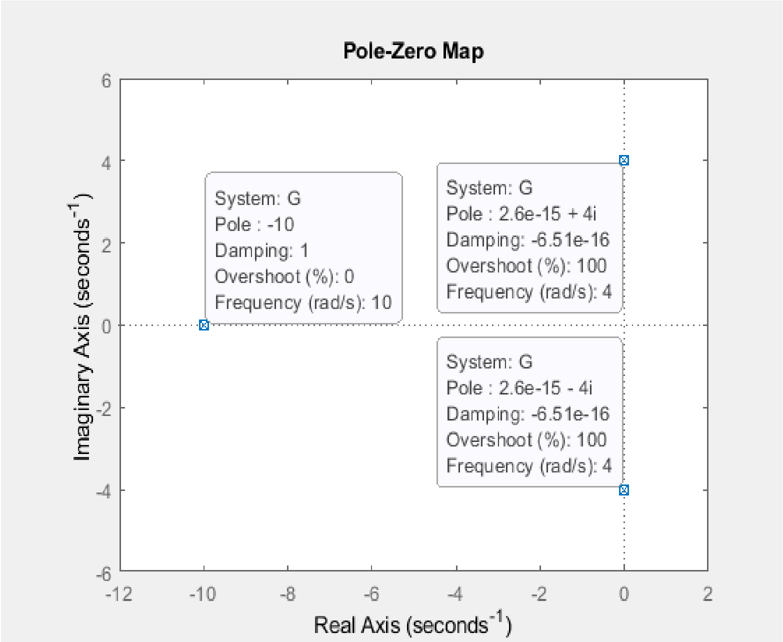
Bounded-input, Bounded-output stability

- **For example,** if the characteristic equation of a closed-loop system is

$$(s+10)\left(s^2+16\right)=0$$







- **The system is said to be marginally stable.**
- **If this system is excited by a sinusoid of frequency ω = 4, the output becomes unbounded.**

Stability Analysis

ROUTH–HURWITZ STABILITY CRITERION

The characteristic equation is written as

$$\Delta(s) = q(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0 = 0$$

To ascertain the stability of the system, it is necessary to determine whether any one of the roots of ***q(s)*** lies in the right half of the s-plane.

Necessary but not sufficient Conditions for stability:

1. All the coefficients of the polynomial have the same sign.
2. All the coefficients be nonzero.

Stability Analysis

ROUTH–HURWITZ STABILITY CRITERION

Example

$$q(s) = (s+2)\left(s^2 - s + 4\right) = \left(s^3 + s^2 + 2s + 8\right).$$

The system is **unstable**, and yet the polynomial possesses **all positive coefficients**.

Stability Analysis

ROUTH–HURWITZ STABILITY CRITERION

- The Routh–Hurwitz criterion is based on ordering the coefficients of the characteristic equation into an array as follows:

$$a_n s^n + a_{n-1} s^{n-1} + a_{n-2}\, s^{n-2} + \cdots + a_1 s + a_0 = 0$$

$$\begin{array}{c|cccc}
s^n & a_n & a_{n-2} & a_{n-4} & \cdots \\
s^{n-1} & a_{n-1} & a_{n-3} & a_{n-5} & \cdots \\
s^{n-2} & b_{n-1} & b_{n-3} & b_{n-5} & \cdots \\
s^{n-3} & c_{n-1} & c_{n-3} & c_{n-5} & \cdots \\
\vdots & \vdots & \vdots & \vdots & \\
s^0 & h_{n-1} & & &
\end{array}$$

$$b_{n-1} = \frac{a_{n-1}a_{n-2} - a_n a_{n-3}}{a_{n-1}} = \frac{-1}{a_{n-1}} \begin{vmatrix} a_n & a_{n-2} \\ a_{n-1} & a_{n-3} \end{vmatrix}$$

$$b_{n-3} = -\frac{1}{a_{n-1}} \begin{vmatrix} a_n & a_{n-4} \\ a_{n-1} & a_{n-5} \end{vmatrix}, \cdots$$

$$c_{n-1} = \frac{-1}{b_{n-1}} \begin{vmatrix} a_{n-1} & a_{n-3} \\ b_{n-1} & b_{n-3} \end{vmatrix}, \cdots$$

Stability Analysis

ROUTH–HURWITZ STABILITY CRITERION

- **The Routh–Hurwitz criterion states** that the **number** of **roots** of q(s) with **positive** real parts is **equal** to the **number** of **changes** in **sign** of the **first column** of the **Routh array**.
- This criterion requires that there be **no changes in sign** in the first column for a **stable system**.
- This requirement is both **necessary** and **sufficient**.
- **Four distinct cases** or configurations of the first column array must be considered, and each must be treated separately and requires suitable modifications

Stability Analysis

ROUTH–HURWITZ STABILITY CRITERION

Case 1. No element in the first column is zero

$$q(s) = a_2 s^2 + a_1 s + a_0$$

The Routh array is written as

$$\begin{array}{c|cc} s^2 & a_2 & a_0 \\ s^1 & a_1 & 0 \\ s^0 & b_1 & 0 \end{array}$$

$$b_1 = \frac{a_1 a_0 - (0) a_2}{a_1} = \frac{-1}{a_1} \begin{vmatrix} a_2 & a_0 \\ a_1 & 0 \end{vmatrix} = a_0$$

Therefore, the requirement for a stable second-order system is that **all** the **coefficients** be **positive,** or all the coefficients be negative.

Stability Analysis

ROUTH–HURWITZ STABILITY CRITERION

Case 1. No element in the first column is zero

$$q(s) = \left(s - 1 + j\sqrt{7}\right)\left(s - 1 - j\sqrt{7}\right)(s + 3) = s^3 + s^2 + 2s + 24$$

The Routh array is written as

$$\begin{array}{c|cc} s^3 & 1 & 2 \\ s^2 & 1 & 24 \\ s^1 & -22 & 0 \\ s^0 & 24 & 0 \end{array}$$

- The polynomial satisfies all the **necessary conditions** because **all** the **coefficients exist** and are **positive**.
- Because **two changes** in **sign** appear in the **first column**, we find that **two roots** of q(s) lie in the **right-hand plane.**

Stability Analysis

ROUTH–HURWITZ STABILITY CRITERION

Case 2. There is a zero in the first column, but some other elements of the row containing the zero in the first column are nonzero

$$q(s) = s^5 + 2s^4 + 2s^3 + 4s^2 + 11s + 10.$$

The Routh array is written as

s^5	1	2	11
s^4	2	4	10
s^3	$\in$	6	0
s^2	c_1	10	0
s^1	d_1	0	0
s^0	10	0	0

$$c_1 = \frac{4 \in -12}{\in}$$

$$d_1 = \frac{6c_1 - 10 \in}{c_1}$$

- If only one element in the array is zero, it may be replaced with a small positive number, $\in$, **When $0 < \in << 1$, we find that $c_1 < 0$ and $d_1 > 0$.**
- Therefore, there are **two sign changes** in the first column; hence the system is **unstable** with **two roots** in the right half-plane.

Stability Analysis

ROUTH–HURWITZ STABILITY CRITERION

Case 3. There is a zero in the first column, and the other elements of the row containing the zero are also zero

- This condition occurs when the polynomial contains singularities that are **symmetrically located** about the **origin** of the s-plane.
- Case 3 occurs when factors such as $(s+\sigma)(s-\sigma)$ or $(s+j\omega)(s-j\omega)$ occur.
- This problem is circumvented by utilizing the **auxiliary polynomial**, ***U(s)***, which immediately precedes the zero entry in the Routh array.
- The **order** of the auxiliary polynomial is always **even** and indicates the **number** of **symmetrical root** pairs.

Stability Analysis

ROUTH–HURWITZ STABILITY CRITERION

Case 3. There is a zero in the first column, and the other elements of the row containing the zero are also zero

Example

$$q(s) = s^3 + 2s^2 + 4s + K$$

$$\begin{array}{c|cc} s^3 & 1 & 4 \\ s^2 & 2 & K \\ s^1 & \dfrac{8-K}{2} & 0 \\ s^0 & K & 0 \end{array}$$

- For a stable system, we require that $0 < K < 8$
- When $K = 8$, we have two roots on the jω-axis and a **marginal stability** case
- we obtain a row of zeros (Case 3) when $K = 8$.
- The **auxiliary polynomial**, ***U(s)***, is the equation of the row **preceding** the row of zeros.

ROUTH–HURWITZ STABILITY CRITERION

Case 3. There is a zero in the first column, and the other elements of the row containing the zero are also zero

Example

$$q(s) = s^3 + 2s^2 + 4s + K$$

$$\begin{array}{c|cc} s^3 & 1 & 4 \\ s^2 & 2 & K \\ s^1 & \dfrac{8-K}{2} & 0 \\ s^0 & K & 0 \end{array}$$

$$U(s) = 2s^2 + Ks^0 = 2s^2 + 8 = 2\left(s^2 + 4\right) = 2(s + j2)(s - j2)$$

- When K = 8, the factors of the characteristic polynomial are

$$q(s) = (s + 2)(s + j2)(s - j2)$$

Stability Analysis

ROUTH–HURWITZ STABILITY CRITERION

Case 4. Repeated roots of the characteristic equation on the jω–axis (Unstable System)

Example $$q(s) = (s+1)(s+j)(s-j)(s+j)(s-j) = s^5 + s^4 + 2s^3 + 2s^2 + s + 1$$

- When $0 < \epsilon << 1$, we note the absence of sign changes in the first column.
- However, as $\epsilon \to 0$, we obtain a row of zero at the s^3 line and a row of zero at the s^1 line.
- The auxiliary polynomial at the s^2 line is $s^2 + 1$, and the auxiliary polynomial at the s^4 line is $s^4 + 2s^2 + 1 = (s^2 + 1)^2$, indicating the repeated roots on the **jω -axis**.
- Hence, the system is **unstable**.

s^5	1	2	1
s^4	1	2	1
s^3	ϵ	ϵ	0
s^2	1	1	
s^1	ϵ	0	
s^0	1		

Stability Analysis

ROUTH–HURWITZ STABILITY CRITERION

Example

$$q(s) = s^5 + s^4 + 4s^3 + 24s^2 + 3s + 63$$

$$U(s) = 21s^2 + 63 = 21\left(s^2 + 3\right) = 21\left(s + j\sqrt{3}\right)\left(s - j\sqrt{3}\right)$$

$$\begin{array}{c|ccc}
s^5 & 1 & 4 & 3 \\
s^4 & 1 & 24 & 63 \\
s^3 & -20 & -60 & 0 \\
s^2 & 21 & 63 & 0 \\
s^1 & 0 & 0 & 0
\end{array}$$

Two roots are on the imaginary axis.

The **two changes** in sign in the first column indicate the presence of **two roots** in the **right-hand** plane, and the system is **unstable**.

Thank You

Any Questions ??